

UNIVERSITY OF TORONTO, ON, Canada

WMO: 715080

Lat: 43.6658N

Lon: 79.3953W

Elev: 113

StdP: 99.98

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.9	-23.4	0.5	-16.0	-19.8	0.7	-12.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	30.8	22.6	29.2	21.8	27.8	21.0	24.2	28.7	23.2	27.4	22.3	26.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.8	26.7	21.9	16.8	25.8	21.0	15.9	25.0	73.4	29.0	69.8	27.4	66.3	26.2	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-20.2	33.5	3.4	2.0	-22.6	35.0	-24.6	36.2	-26.5	37.3	-29.0	38.8		
(5)			-20.8	25.7	3.2	1.5	-23.1	26.8	-24.9	27.7	-26.7	28.5	-29.1	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.7	-3.0	1.5	7.8	14.6	19.7	22.7	22.1	18.4	11.3	5.3	-0.1	(6)
(7)		DBStd	10.34	6.09	5.35	5.50	4.51	4.33	3.45	2.85	2.70	3.64	4.22	4.35	4.64	(7)
(8)		HDD10.0	1652	424	364	267	92	8	0	0	0	0	34	149	313	(8)
(9)		HDD18.3	3506	682	597	521	315	133	25	1	3	43	222	392	572	(9)
(10)		CDD10.0	1576	0	0	4	28	150	291	395	375	252	74	7	0	(10)
(11)		CDD18.3	387	0	0	0	0	17	65	138	119	45	3	0	0	(11)
(12)		CDH23.3	2369	0	0	0	2	104	394	958	683	217	10	0	0	(12)
(13)		CDH26.7	580	0	0	0	0	19	97	270	149	43	2	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	847	61	56	65	69	72	69	74	82	77	63	78	76	(15)
(16)		PrecMax	1158	123	98	155	125	150	151	142	196	222	142	190	154	(16)
(17)		PrecMin	648	21	20	13	30	26	17	20	17	10	12	13	24	(17)
(18)		PrecStd	119	27	26	32	24	30	33	31	44	45	32	36	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	12.3	17.9	22.6	28.9	31.6	33.6	32.0	30.4	24.4	16.9	11.9	(19)
(20)			MCWB	10.5	7.8	13.5	14.8	20.7	22.9	24.1	23.5	22.1	19.4	13.3	10.5	(20)
(21)		2%	DB	8.7	8.2	13.9	19.3	26.2	29.3	31.2	29.9	27.7	21.5	14.6	9.3	(21)
(22)			MCWB	7.4	5.8	10.3	12.7	18.8	21.4	22.8	22.8	21.4	17.4	11.6	7.7	(22)
(23)		5%	DB	6.2	5.5	11.1	16.8	23.9	27.4	29.5	28.2	25.7	19.4	12.9	7.5	(23)
(24)			MCWB	4.7	3.5	8.0	10.7	17.2	20.3	21.8	21.5	20.0	15.9	10.3	5.9	(24)
(25)		10%	DB	4.1	3.7	8.8	14.6	21.5	25.6	27.9	26.8	24.1	17.5	11.4	5.8	(25)
(26)			MCWB	2.8	2.0	5.9	9.5	15.5	19.3	21.0	20.6	19.4	14.6	9.2	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	8.9	14.4	16.3	22.2	24.2	25.9	25.5	23.7	20.7	14.0	10.6	(27)
(28)			MCD B	11.6	10.4	16.7	20.3	26.9	29.8	31.4	29.6	28.1	23.6	16.1	11.6	(28)
(29)		2%	WB	7.5	6.0	11.0	14.0	20.1	22.9	24.4	23.9	22.4	18.1	12.3	8.0	(29)
(30)			MCD B	8.3	8.1	12.9	18.4	25.2	27.3	29.2	27.9	26.1	20.6	14.0	9.2	(30)
(31)		5%	WB	4.8	3.7	8.3	11.9	18.2	21.6	23.4	22.9	21.4	16.5	10.7	6.2	(31)
(32)			MCD B	5.8	5.2	10.7	15.7	22.4	25.9	27.7	26.7	24.8	18.6	12.4	7.4	(32)
(33)		10%	WB	2.9	2.1	6.0	9.9	16.4	20.3	22.4	21.9	20.2	15.1	9.3	4.3	(33)
(34)			MCD B	4.0	3.6	8.3	13.4	20.3	24.1	26.2	25.3	23.1	17.2	11.1	5.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.2	6.6	7.5	8.2	8.2	7.8	7.6	6.4	5.6	5.1	(35)	
(36)			MCDBR	7.1	8.1	9.4	11.1	11.1	10.5	10.0	9.4	9.7	8.7	7.8	6.4	(36)
(37)			MCWBR	6.8	7.0	7.1	7.5	6.3	5.7	4.9	4.6	5.0	6.0	6.4	6.1	(37)
(38)		5% WB	MCD BR	6.8	7.9	9.2	10.6	10.1	9.6	8.8	8.3	8.4	8.0	7.2	6.4	(38)
(39)			MCWBR	6.9	7.1	7.5	8.2	6.4	6.0	4.9	4.7	5.1	6.0	6.5	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.321	0.334	0.382	0.412	0.440	0.434	0.420	0.387	0.366	0.321	0.302	(40)	
(41)		taud	2.400	2.339	2.399	2.302	2.270	2.220	2.262	2.289	2.359	2.394	2.483	2.440	(41)	
(42)		Ebn at Noon	827	869	905	882	862	835	836	835	836	806	800	790	(42)	
(43)		Edh at Noon	76	96	104	124	132	139	132	124	107	91	71	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.15	3.44	4.43	5.38	5.84	5.87	5.16	4.13	2.48	1.55	1.13	(44)	
(45)		RadStd	0.13	0.20	0.34	0.47	0.57	0.43	0.43	0.26	0.38	0.22	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-1.32	-1.56	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page